



BEST PRACTICE SERIES

Fuel Consumption / Large
Track-Type Tractor
Production Study

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

Fuel Consumption is an important element of the cost per ton / cost per yard equation. This best practice deals with measuring the fuel consumption of a Large Track-Type Tractor during a production study. It also suggests guidelines for measuring the time, volume and centroidal push distance of a track-type tractor productivity study.

2.0 Best Practice Description

This best practice recommends measuring fuel consumption by weight rather than by volume. It recommends completely draining the tractor's fuel tank before the study, and then transferring a weighed, measured amount of fuel from an auxiliary fuel tank to the tractor's fuel tank. Upon completion of the production study, the remaining fuel is transferred back to the auxiliary fuel tank and weighed. The weight of the fuel can then be converted into liters/gallons.



Above left, a 250-gallon plastic tote is utilized for an auxiliary fuel tank in which the fuel is transferred into the tractor by gravity feed. Fuel is drained out of the tractor by gravity feed.

Middle, 55-gallon steel drums are utilized for auxiliary fuel tanks in which the transfer of fuel into the tractor occurs by an electric transfer pump. Fuel is drained out of the tractor by gravity feed. Above right, technician reads scale weight of barrel of diesel fuel.

Key Elements of A Productivity Study:

1. Time
2. Volume
3. Distance
4. Machine Health
5. Operators
6. Pit Study vs. Area Study

2.1 Time

- Time can be measured as a single segment from start to finish or it can be measured in cycle segments such as doze time, carry time, out-of-pit time and reverse time.
- It is recommended to use a digital stopwatch displaying hundredths of a minute.
- A study can be timed in minutes and seconds, but the key is not mix the two type of measurement.

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Production Study

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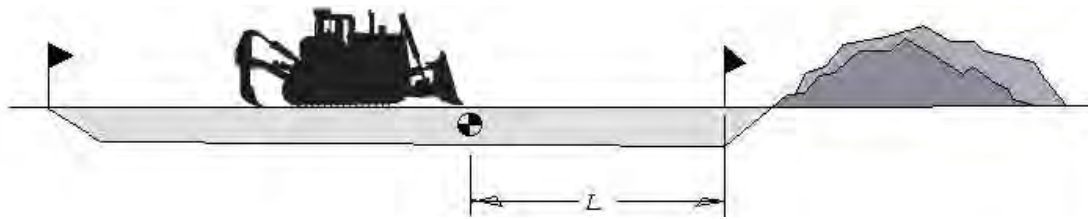
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2.2 Volume

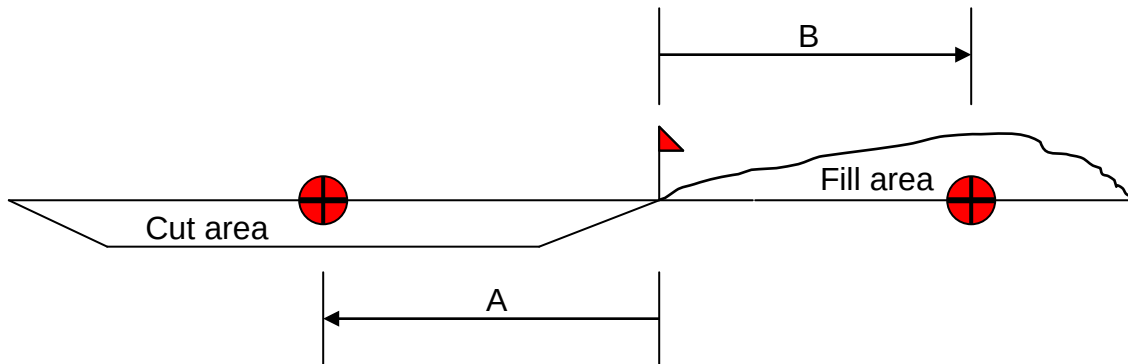
- Volume is measured by means of a survey taken before and after a test.
- Recommend the customer or a third party conduct the survey as it makes the study more creditable.
- In a non-fuel production study, the cut area volume and corresponding average push distance, (centroidal push distance), is all that is required.
- In a fuel production study, the cut area volume and corresponding centroidal push distance is required as well as the centroidal push distance for the fill area.
- It is desirable to match the fill area volume with the cut area volume, but in reality is difficult to achieve.

2.3 Distance

- Production is a function of the cycle time and blade load capabilities. One of the variables affecting cycle time is the distance the blade load of material is moved. If two tractors move material the same distance, cycle times, as a function of each tractors design and capability, can be compared and analyzed. However, if the doze distances are not the same for the other tractor, then cycle times are not directly comparable, nor are the rates of production.
- In a production study, it is possible through surveying to not only determine the pit volume but also the location of the pit centroid – as defined by the centroidal push distance “L”. The centroid is also defined as the center of mass of the material moved.

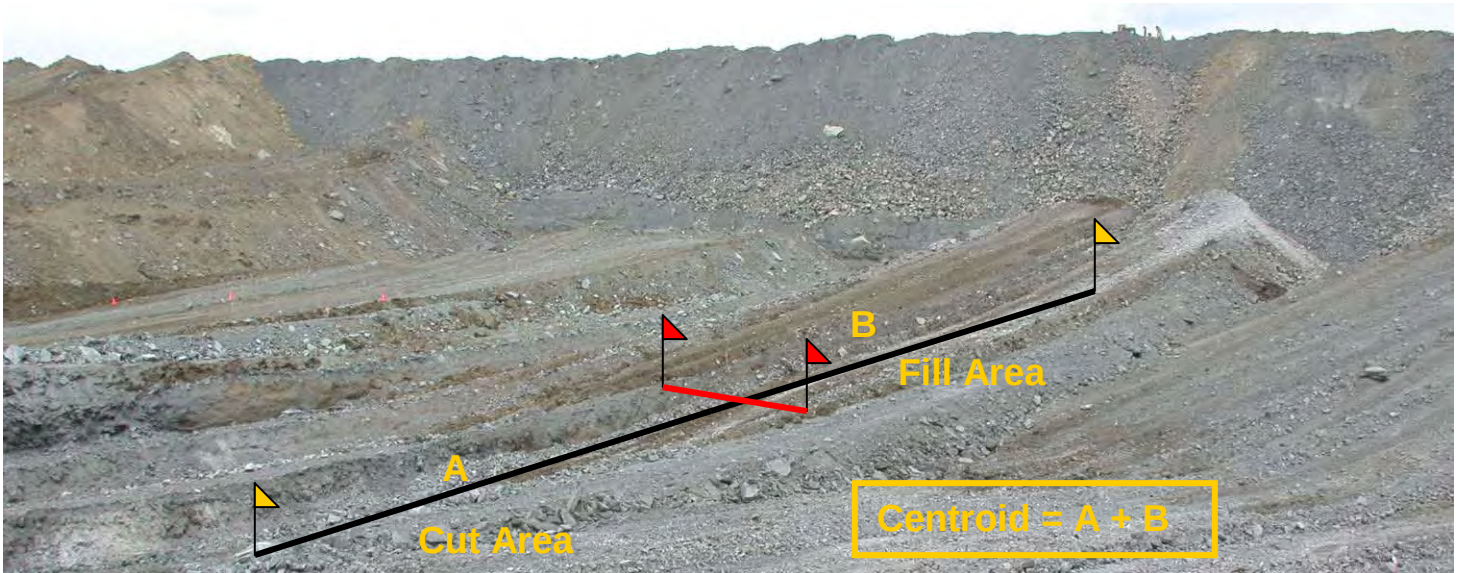


- When measuring fuel consumption during a productivity study, it is important to establish a centroidal push distance for both the cut area and the fill area. Combining the two centroidal distances provides an overall centroidal distance for that pit. Caterpillar Competitive Report TEXR 0524 illustrates how centroidal distances are applied to a Productivity Study. This report can be found in the Global Sales Library.



Centroidal Distance = Distance A + Distance B

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The above picture illustrates the measurement of a centroidal push distance for a test pit of a track-type tractor fuel / production study.

2.4 Machine Health

- Before the study, carefully select tractors of equal comparison. Key comparisons include blade capacities; ripper or counterweight equipped, track width and grouser height.
- Make certain machines are within specifications. Be sure to account for the following items:
 - Torque Converter Stall
 - High Idle
 - Low Idle
 - Boost pressure
 - Fuel Filters
 - Air Filters
 - Ground Engaging Tools
 - Cutting Edges ... new vs. old and thick vs. thin, as it can effect penetration
 - Ripper Tooth ... new
 - Grouser Height ... affects traction and productivity

2.5 Operator

- Operator technique can have an influence on the outcome of the study if more than one operator is used.
- Conducting a series of multiple pits and switching operators and tractor location within the pits will produce the highest level of confidence for the study results.
- Before the testing begins, adequate time is needed for the operators to become familiar with tractors they are requested to operate.

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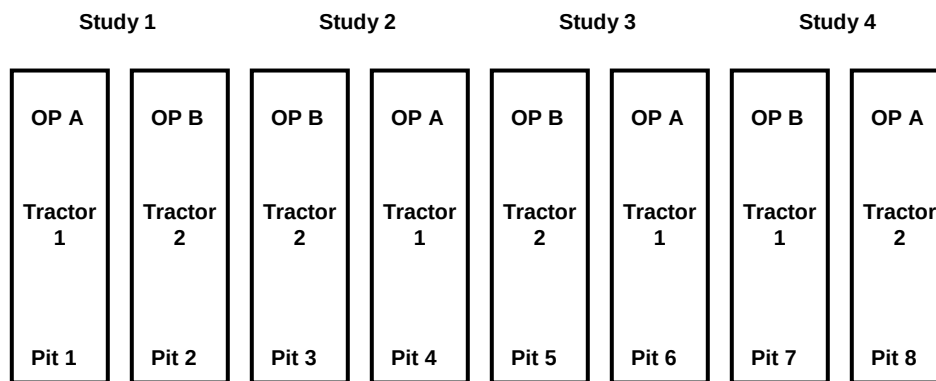
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2.6 Pit Study vs. Area Study

- The best method to measure Track-Type Tractor productivity is to use a series of single slots.
- The single slot reduces the number of variables introduced to the study.
- An area dozing study is a valid study but introduces more variability due to operator technique, especially when removing the center berm of material left between slots.
- For Width of a single pit, stake out an area that is approximately one and a half times the width of the tractor. Multiple single pits should be utilized in the total study.
- For Length, stake out an area that is long enough to accommodate one hour of study time. The length will also vary according to material type and tractor size. 200 feet is a good distance to start from.
- For Depth, dig to a depth of approximately 75% to 100% of blade height.



- The above diagram illustrates the layout of a multiple pit tractor study, switching operators and tractor location in each study.

3.0 Implementation Steps

The first step is to estimate the amount of fuel needed to conduct the study. Too little fuel and the tractor will run out of fuel during the study. Too much fuel and it will take longer to transfer the fuel in and out of the auxiliary tank at the beginning and end of the study. Obtain an adequate sized auxiliary fuel tank, large enough to hold the amount of fuel to be used in the study. The auxiliary tank may range from a metal barrel to a plastic tote.

The transfer of fuel is accomplished by either a transfer pump or gravity fed. If gravity fed, then a lifting device such as a service truck equipped with a boom is required. A load link or scale, capable of weighing the amount of fuel to be used during the study, is also required.

4.0 Benefits

Measuring fuel consumption by weight is more accurate and reliable than measuring it by volume or the use of fuel flow meters. Topping off the fuel tank before and after a study is a simple method, but is highly dependent on many factors and therefore is not always accurate. The following factors contribute to a low level of confidence when measuring fuel consumption by topping off the fuel tank. The tractor must be parked on a perfectly level location for both fueling and refueling. One must be assured that no voids or air bubbles remain in the tank during fueling. The fuel tank must always be filled to the exact same level, before and after the test. This process requires a calibrated fuel meter for measuring the amount of fuel pumped into the tractor's fuel tank. This process requires

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compensating for the difference of fuel temperature before and after a test, as hot fuel returned from the engine to the fuel tank will affect the volume of the fuel in the tank.

Fuel flow meters mounted on Track-Type Tractors in the past have not proven to be reliable. It requires two flow meters per tractor, one for the fuel supply line and one for the fuel return line. It requires splicing into the fuel lines, which is messy and is difficult to return the fuel lines back to original condition after the study, especially on competitive tractors.

Using auxiliary tanks to measure fuel consumption by weight has been successful with small and medium size track-type tractors by attaching the auxiliary tank to the tractor. The difficulty of this method for large track-type tractors is finding an adequate location to mount a tank large enough to complete a study. The tractor's fuel lines also need to be spliced.

Caterpillar Large Track-Type Tractors today are equipped with an ECM (Electronic Control Module) that records total fuel burned. The ECM estimates fuel consumption rather than actually measuring fuel consumption. Also, competitive tractors may not have a comparable onboard fuel burn measurement system.

5.0 Resources Required

- Auxiliary fuel tanks, one per tractor. Auxiliary tanks may be in the form of metal barrels or plastic totes.
- Fuel Transfer pumps if transferring fuel from barrels.
- Appropriate diameter and length hoses for fuel transfer.
- Lifting straps.
- Service truck equipped with a crane or lifting device.
- Load Link or scale capable of weighing amount of fuel needed for the study.

6.0 Supporting Attachments / References

Caterpillar D9R vs. Komatsu D275A-5 Competitive Report TEJB 9563

Caterpillar D11T vs. Komatsu D475A-5 Competitive Report TEXR 0524

7.0 Related Best Practices

0808-2.15-1129 - Managing Fuel Consumption Reduces Cost

8.0 Acknowledgements

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